194754: metamonzogranite, Deep Well

(Walganna Suite, Bald Rock Supersuite, Youanmi Terrane, Yilgarn Craton)

Location and sampling

Ninghan (SH 50-7), Maranalgo (2439)
MGA Zone 50, 564433E 6772662N

Sampled on 15 May 2009

This sample was collected from outcrop northeast of the Paynes Find – Yalgoo Road on Oudabunna Station, about 5.5 km south of Bullamania Well, 4.2 km west of the Great Northern Highway, and 3.2 km northeast of Deep Well.

Geological context

The unit sampled is the 2650–2600 Ma Walganna Suite of the post-tectonic Bald Rock Supersuite (Van Kranendonk et al., 2013). The Walganna Suite consists mainly of non-deformed, coarse-grained, equigranular or K-feldspar-porphyritic monzogranite to syenogranite, and forms about 40% of the bedrock in the western Youanmi Terrane (Zibra, 2021b). Bald Rock Supersuite granitic rocks locally crosscut foliated 2665–2640 Ma Jungar Suite and 2697–2657 Ma Tuckanarra Suite granites, all Murchison Supergroup supracrustal rocks, and all granitic and mafic-ultramafic rocks of the Annean and Austin Downs Supersuites (Zibra, 2021a). A monzogranite of the Walganna Suite collected about 51 km to the west yielded an igneous crystallization age of 2626 ± 6 Ma (GSWA 207630, Wingate et al., 2014). A metagranodiorite collected about 1.0 km to the west-northwest yielded an igneous crystallization age of 2710 ± 4 Ma (GSWA 201514, Wingate et al., 2021).

Petrographic description

The sample is a coarse-grained metamonzogranite (Fig. 1), consisting of about 40% quartz, 35% K-feldspar, 20% plagioclase, 3% biotite, <1% allanite, and trace magnetite and zircon. Quartz is anhedral, irregular, up to 10 mm in size, and weakly strained. K-feldspar is anhedral to subhedral, tabular, up to 16 mm long, simply twinned, and contains abundant plagioclase and minor quartz inclusions. K-feldspar is microperthitic and there is some development of fine microcline twinning. Plagioclase (andesine, An_{36}) is subhedral, mainly equant, up to 7 mm in diameter, and exhibits weak zoning and minor alteration to sericite and clay minerals. Incipient myrmekite is developed at some boundaries between plagioclase and K-feldspar. Biotite forms subhedral, slightly ragged laths up to 3.5 mm long, locally altered to chlorite, and metamict allanite forms subhedral to euhedral, tabular to rhombic grains up to 1.5 mm long. Magnetite is anhedral and up to 0.7 mm in diameter. The development of microcline twinning and rare myrmekite suggests that the sample has undergone some solid-state deformation at low to moderate metamorphic grade.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown, and subhedral to euhedral. The crystals are up to 350 µm long, and mostly elongate, with aspect ratios up to 8:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and crystals contain cores that exhibit evidence of dissolution. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 19–20 July 2019, using SHRIMP-A, and 23 April 2020, using SHRIMP-B. Analyses 1.1 to 18.1 (spot numbers 1–20) were obtained during the first session, together with 11 analyses of the CZ3 standard, of which 10 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.23% (1σ). Analyses 19.1 to 21.1 (spot numbers 21–24) were obtained during the second session, together with three analyses of the CZ3 standard, which indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.43% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-four analyses were obtained from 21 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3), and form a single group, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 24 analyses of 21 zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2627 ± 3 Ma (MSWD = 1.2).

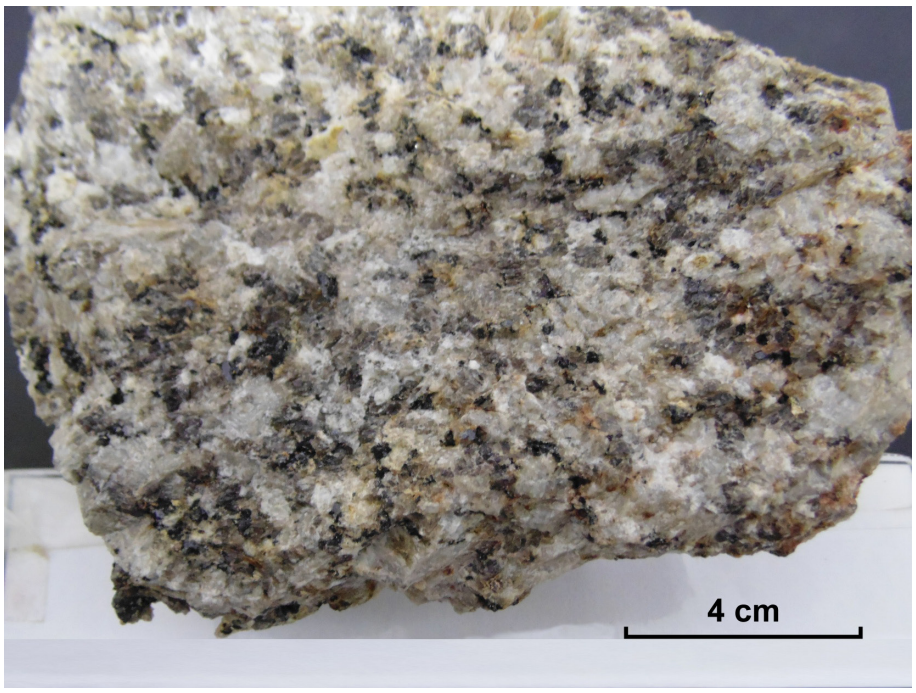

Figure 1. Hand specimen image of sample 194754: metamonzogranite, Deep Well

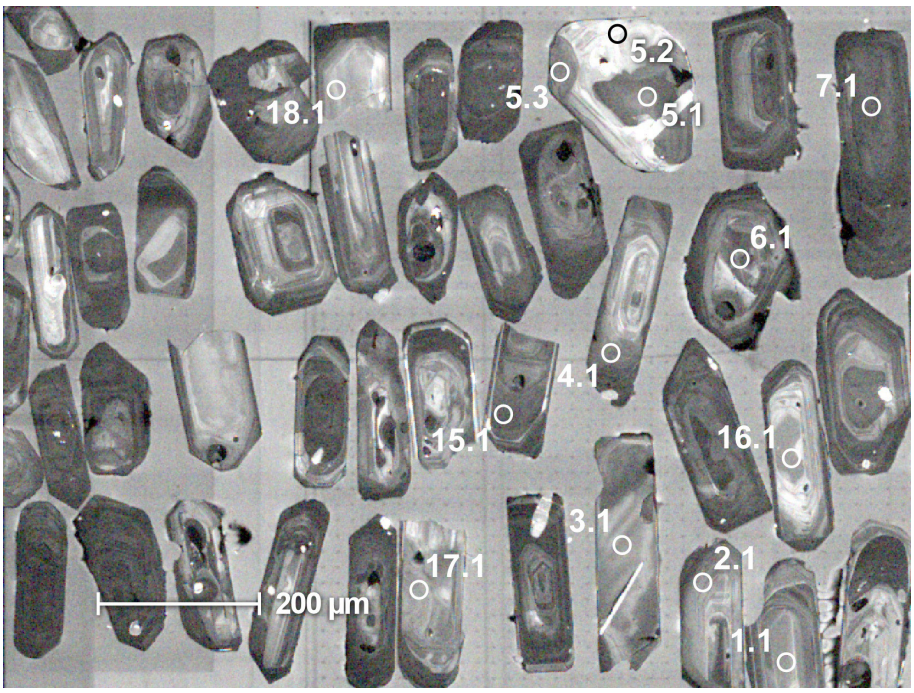

Figure 2. Cathodoluminescence image of representative zircons from sample 194754: metamonzogranite, Deep Well. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 194754: metamonzogranite, Deep Well

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
I	13	11.1	49	64	1.29	0.060	1.982	0.028	0.17582	0.00142	1.983	0.028	0.17528	0.00147	2633	31	2609	14	-0.9
I	7	5.3	101	60	0.60	0.029	1.990	0.021	0.17560	0.00098	1.990	0.021	0.17534	0.00100	2625	23	2609	9	-0.6
I	16	14.1	160	97	0.60	0.052	2.044	0.018	0.17646	0.00077	2.045	0.018	0.17599	0.00079	2567	19	2615	7	1.9
I	19	17.1	69	44	0.65	0.064	2.016	0.027	0.17665	0.00121	2.017	0.027	0.17608	0.00125	2596	28	2616	12	0.8
I	4	4.1	106	79	0.75	-0.014	1.999	0.021	0.17598	0.00095	1.998	0.021	0.17610	0.00096	2616	23	2617	9	0.0
I	3	3.1	69	60	0.88	0.131	2.025	0.025	0.17731	0.00121	2.027	0.025	0.17614	0.00131	2585	27	2617	12	1.2
I	8	6.1	213	530	2.49	0.007	2.049	0.017	0.17636	0.00070	2.049	0.017	0.17630	0.00070	2562	18	2618	7	2.1
I	1	1.1	79	79	1.01	0.019	2.013	0.024	0.17654	0.00114	2.013	0.024	0.17637	0.00116	2600	25	2619	11	0.7
I	5	5.1	151	99	0.65	0.000	2.047	0.029	0.17643	0.00121	2.047	0.029	0.17643	0.00121	2564	30	2620	11	2.1
I	6	5.2	38	35	0.92	0.123	1.955	0.052	0.17762	0.00167	1.958	0.052	0.17651	0.00179	2660	58	2620	17	-1.5
I	23	17.2	374	230	0.61	0.039	2.054	0.018	0.17709	0.00054	2.054	0.018	0.17675	0.00056	2557	19	2623	5	2.5
I	21	19.1	534	201	0.38	0.077	2.220	0.025	0.17771	0.00048	2.221	0.025	0.17702	0.00050	2396	23	2625	5	8.7
I	18	16.1	59	62	1.05	0.025	2.060	0.027	0.17725	0.00131	2.061	0.027	0.17703	0.00133	2550	28	2625	12	2.9
I	17	15.1	287	156	0.54	0.053	2.055	0.016	0.17755	0.00057	2.056	0.016	0.17707	0.00059	2555	17	2626	6	2.7
I	2	2.1	150	76	0.51	0.069	2.057	0.020	0.17772	0.00090	2.058	0.020	0.17711	0.00093	2553	20	2626	9	2.8
I	24	21.1	251	94	0.37	0.000	2.087	0.019	0.17734	0.00135	2.087	0.019	0.17734	0.00135	2524	20	2628	13	4.0
I	10	8.1	91	63	0.69	0.045	2.055	0.023	0.17774	0.00102	2.056	0.023	0.17734	0.00104	2555	23	2628	10	2.8
I	9	7.1	1447	951	0.66	0.005	1.996	0.012	0.17765	0.00026	1.996	0.012	0.17761	0.00026	2618	14	2631	2	0.5
I	20	18.1	53	72	1.36	-0.026	2.031	0.027	0.17741	0.00134	2.031	0.027	0.17765	0.00136	2581	29	2631	13	1.9
I	11	9.1	69	65	0.94	0.120	2.145	0.027	0.17872	0.00128	2.147	0.027	0.17765	0.00137	2465	26	2631	13	6.3
I	14	12.1	32	29	0.92	-0.265	1.997	0.033	0.17535	0.00171	1.992	0.033	0.17771	0.00196	2622	36	2632	18	0.4
I	12	10.1	35	30	0.84	-0.042	2.070	0.034	0.17734	0.00169	2.069	0.034	0.17771	0.00173	2542	34	2632	16	3.4
I	22	20.1	405	138	0.34	0.023	2.055	0.017	0.17885	0.00052	2.055	0.017	0.17865	0.00053	2556	19	2640	5	3.2
I	15	13.1	87	105	1.21	-0.137	2.088	0.023	0.17825	0.00109	2.085	0.023	0.17947	0.00117	2526	24	2648	11	4.6

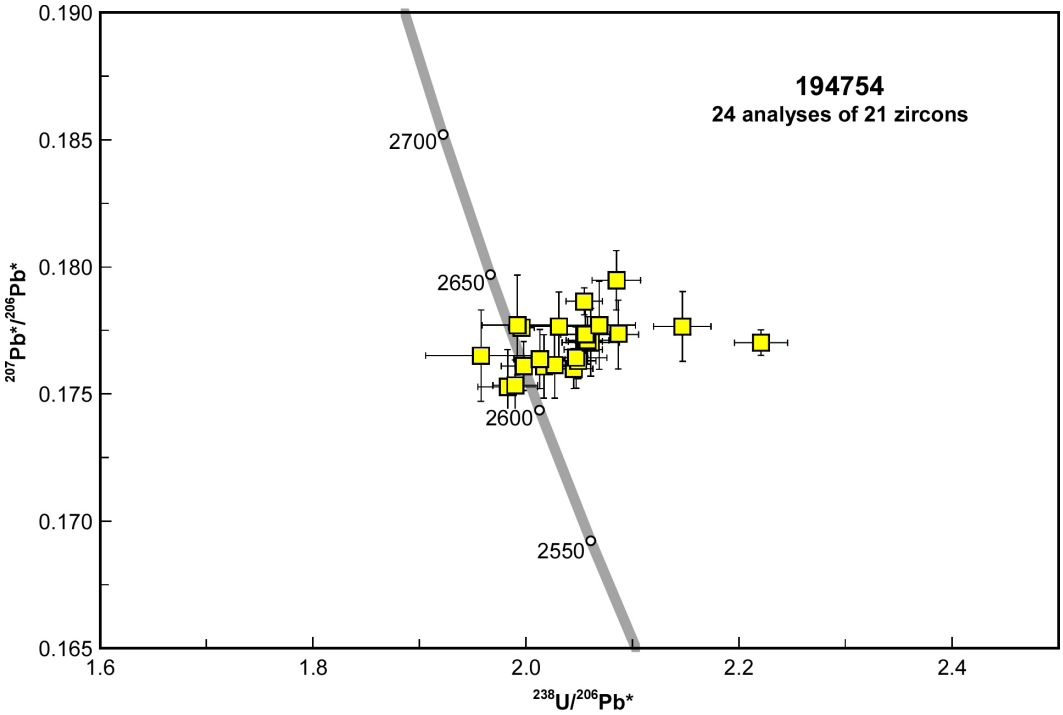

Figure 3. U–Pb analytical data for sample 194754: metamonzogranite, Deep Well. Yellow squares indicate Group I (magmatic zircons)

The date of 2627 ± 3 Ma for the 24 analyses in Group I is interpreted as the igneous crystallization age of the monzogranite. No significant or systematic difference in age could be discerned between the cores and the outer parts of the zircons.

References

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Ivanic, T 2021, 194754: metamonzogranite, Deep Well; Geochronology Record 1821: Geological Survey of Western Australia, 4p.

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